

Relation Algebra

§8. / is all of

(8a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/compares images: $S(y) \subseteq R(x)$. \ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.

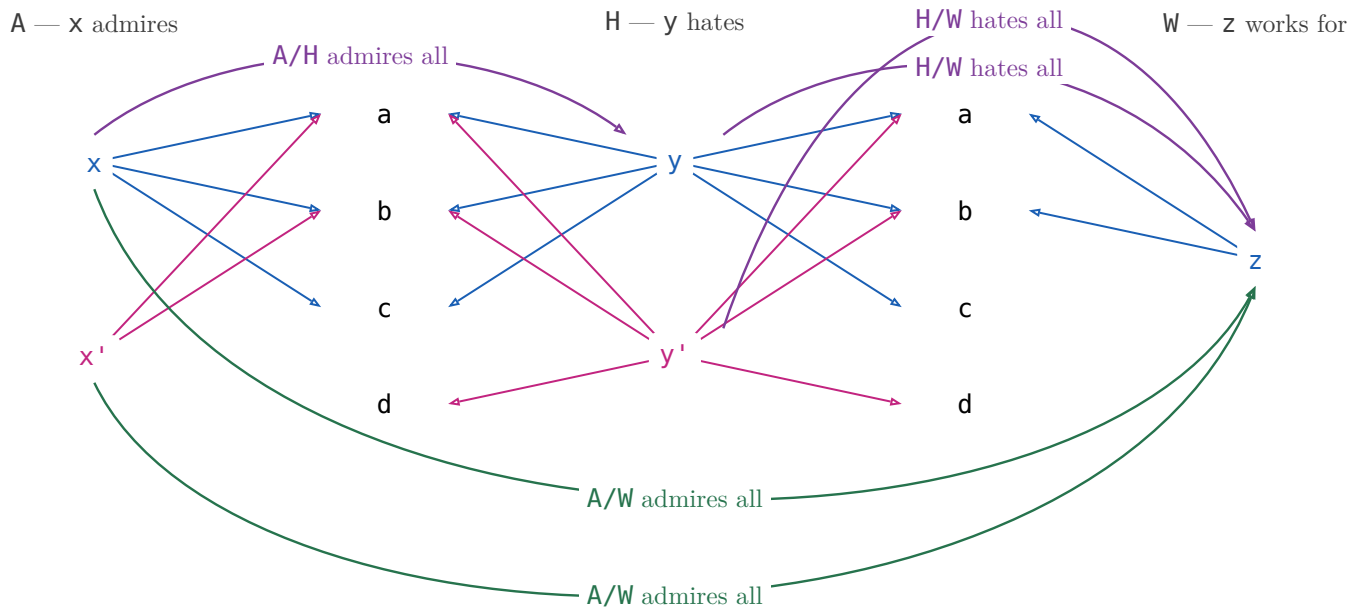
example: A admires, H hates, W works for

$x (A/H) y$ — x admires everyone y hates.

$x (H \setminus A) y$ — everyone who hates x admires y .

§8.1. $(R/S)(S/W) \sqsubseteq R/W$

(8.1a)



(8.1b)

$$x (A/H) y \text{ — } x \text{ admires everyone } y \text{ hates}$$

$$y (H/W) z \text{ — } y \text{ hates everyone } z \text{ works for}$$

$$x (A/W) z \text{ — } x \text{ admires everyone } z \text{ works for}$$

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \sqsubseteq R/W$.

$X \in R/S \iff XS \in R$ X is any x -to- y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
$X \in S \setminus R \iff SX \in R$ The mirror — divide on the left when x comes first.	
$(R/S)S \in R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S=\emptyset$: R/S is everyone, $(R/S)S=\emptyset$.	
$S \setminus (S \setminus R) \in R$ The mirror.	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
maps: $f \setminus (R/S) = (fR)/S$ Rename x before or after dividing — the licence to write fR/S .	
$R/(fS) = (R/S) f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
$(R/S)(S/W) \in R/W$ Someone who admires all of a hate-set that already covers everyone z works for admires those people too.	
$1 \in R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
$(R/R)(R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \in R/R$ above it is an equality.	
$(R/R)R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-set is admiring each set in full.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: $\cap$, $\cup$, $\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.